

Prompt Engineering and Generative AI - Fundamentals

How to run the notebooks in Google Colab (October 2026)

Course: [udemy.com/course/prompt-engineering-and-generative-ai-fundamentals](https://www.udemy.com/course/prompt-engineering-and-generative-ai-fundamentals)

GitHub: github.com/jsathish1990/prompt-engineering-and-generative-ai-fundamentals

1. Before you start

- The first cell in each notebook installs the library versions it needs. No code changes are needed.
- The library versions were tested in October 2026 on Python 3.12, the version Colab uses.
- The notebooks use the OpenAI and Google Gemini APIs. API usage is charged to your own account.

2. API keys

Key	Where to get it	Used in
OPENAI_API_KEY	platform.openai.com → API keys	Most notebooks
GOOGLE_API_KEY	aistudio.google.com → Get API key	Gemini notebooks
HF_TOKEN	huggingface.co → Settings → Access Tokens	Fine-tuning notebooks
LangSmith API key	smith.langchain.com → Settings → API keys	RAGAS_langsmith

- In Colab, click the **key icon** (Secrets) on the left → **Add new secret** → use the name above → turn on **Notebook access**.
- PyPDF_RAG_with_RAGAS and RAGAS_langsmith set a key inside a cell. Replace the placeholder with your own key.

3. Data files

Three notebooks read files from `/content/drive/MyDrive/`. Copy these files to your Google Drive (My Drive):

File	Where to get it	Used in
data_QA_.csv	Section 2 folder in the GitHub repository	Retrieval_Augmented_Generation_1
ref_.pdf	Download the PDF from arxiv.org/abs/2401.12379 and rename it	PyPDF_RAG_with_RAGAS
Gemma_Google.pdf	Save blog.google/technology/developers/gemma-open-models as a PDF	Guardrails_with_ChatModel

- The two PDFs are not in the repository. Download them from the links above.
- Allow access to your Google Drive when the notebook runs `drive.mount(...)`.

4. Running a notebook

- Open the notebook in Colab: **File** → **Open notebook** → **GitHub**, then paste the repository link.
- Fine-tuning notebooks: **Runtime** → **Change runtime type** → **GPU**. The 7B models need an A100 or L4 GPU and may run out of memory on a T4. Some models need you to accept their terms on Hugging Face first (for example mistralai/Mistral-7B-v0.1).
- Click **Runtime** → **Run all**.
- If Colab asks to restart the session after the install cell, restart and run all again. This happens in the two RAGAS notebooks.
- Pip warnings can be ignored unless a cell fails.

5. Notebooks

Section	Segment	Notebook	Keys
Prompt Engineering	1-4: Definition, best practices, streaming, temperature and tokens, zero-shot, few-shot, chain-of-thought	Set_up_Gemini+(1).ipynb	Google
Prompt Engineering	Setting up the GPT model	Set_up_OpenAI.ipynb	OpenAI
Prompt Engineering	Tree-of-Thoughts, 4x4 Sudoku	4x4_example_OpenAI.ipynb	OpenAI
RAG	1: RAG on a CSV file	Retrieval_Augmented_Generation_1.ipynb	OpenAI
RAG	2: Arxiv loader, FAISS, conversational retrieval chain	Arxiv_Loader_RAG_.ipynb	OpenAI
RAG	3: Evaluation using RAGAS	PyPDF_RAG_with_RAGAS.ipynb	OpenAI
RAG	4: LangSmith with RAGAS	RAGAS_langsmith.ipynb	OpenAI, LangSmith
RAG	5: Gemini embeddings and document search	Document_Q_A_Gemini.ipynb	Google
Fine-tuning	1: Prompting vs fine-tuning	Finetuning_vs_Prompting.ipynb	HF (GPU)
Fine-tuning	2-3: Types of fine-tuning, EDA, task-specific fine-tuning	Fine_tuning_an_LLM_.ipynb	HF (GPU)
Guardrails	1: Guardrails from OpenAI	Topical_guardrail.ipynb	OpenAI
Guardrails	2: GuardrailsAI for extracting information	GuardrailsAI_extracting_text.ipynb	OpenAI
Guardrails	3: GuardrailsAI for generating structured data	GuardrailsAI_generating_structured_data.ipynb	OpenAI
Guardrails	3: GuardrailsAI with a chat model	Guardrails_with_ChatModel.ipynb	OpenAI

6. Changes made in October 2026

- Library versions are pinned in one install line. The old install lines are kept as comments.
- Retired models were replaced. OpenAI gpt-3.5-turbo, gpt-3.5-turbo-instruct and gpt-4 now use **gpt-4o-mini**. Gemini-Pro now uses **gemini-3.1-flash-lite**, and the Gemini embedding model is **models/gemini-embedding-001**.
- 4x4_example_OpenAI, Arxiv_Loader_RAG_ and Retrieval_Augmented_Generation_1 now use the OpenAI chat model (from `langchain_openai import ChatOpenAI as OpenAI`), since the completion model is retired.
- The GuardrailsAI notebooks use `openai.ChatCompletion.create` instead of `openai.Completion.create` for the same reason.
- The RAGAS notebooks have one extra line that sets the RAGAS evaluation model to gpt-4o-mini.
- faiss-gpu was changed to faiss-cpu, since faiss-gpu does not install on Python 3.12.
- The responses will not exactly match the outputs saved in the notebooks.

7. If a notebook stops working

- If a model is retired, replace the model name in the notebook (for example "gpt-4o-mini") with a current model.
- If an import fails, restart the session and run all again.